Run report: one invariant guess per link at m = 4, degree 3, k = 10

- **Study:** `research/dreg_m4_link_guess_d3_20261106/`, pre-registered before any study run. One amendment: the runner's F4 caps, before any outcome was read.
- **Run:** 2026-11-06 16:33 UTC to 17:13 UTC.
- **Class:** stage diagnostic. No `S`, no rho ratio, no speed claim.
- **Standing:** unverified assertion until a pcc verdict exists.

What ran

- **Chain:** $S_3(x_1, x_2, e_1)$, $S_3(e_1, x_3, e_2)$, $S_3(e_2, x_4, x(R))$ at $\ell = 5$, n = 11 and 12. Every committed unsatisfiable draw: 8 in all.
- **Arm I:** four pair sums on $x_1 + x_2$, three guesses on $e_1 + x_3$ and three on $e_2 + x_4$. The link coordinates are the three largest k with $|T_k| \geq 2$ on the draw.
- **Arm R:** ten raw guesses in the same places (top coordinates of x_2, x_3, x_4).
- All 1024 branches of both arms on every draw: 16,384 runs of `dreg_linear_guess --to 3 --check`, plus the identity control N.

Completion

- Every run left a record; none was killed.
- 32 to 34 variables occur after the guesses. The degree-3 closure took 0.002–0.51 s per branch.
- The first attempt (runner without the F4 caps) is kept, unread, in `runs/attempts/run1-no-f4-env-20261106/`.

Measured values

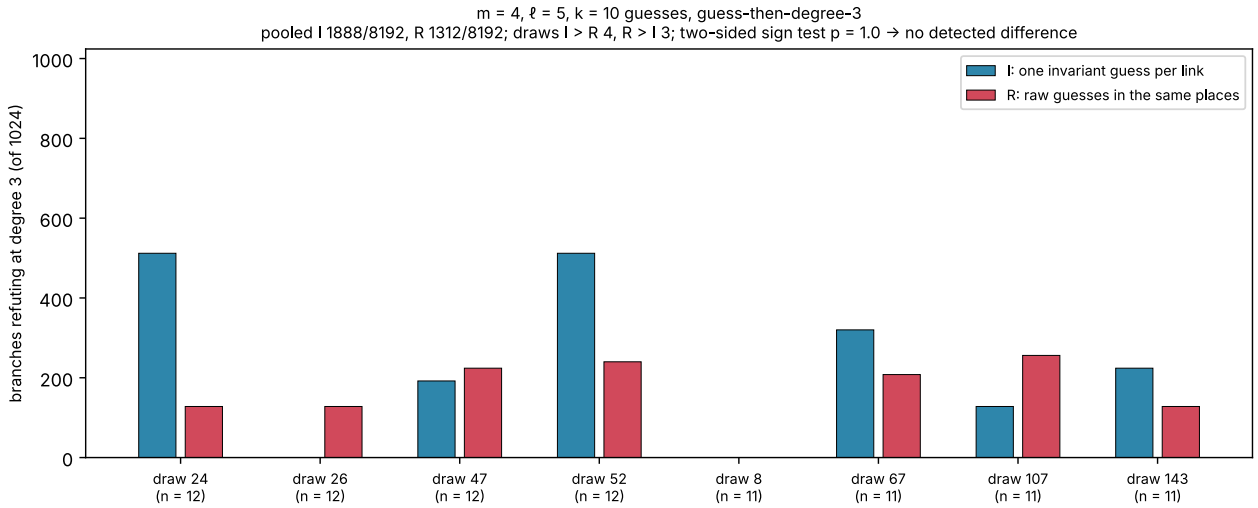

Figure 1: Branches refuting at degree 3, per arm and draw. Source: `plot.py` (layout only) on `plot-data.json` from `dreg_m4_link_score --plot-data`.

- **Controls:** valid. N matches the committed m = 3 record, every record passes the identity and exactness checks with arm I's link guesses re-derived from the basis, and all branches are present.
- **Pooled:** I refutes on 1,888 of 8,192 branches, R on 1,312.
- **By draw:** I > R on 4 draws, R > I on 3, equal (both 0) on 1.
- **Registered verdict:** no detected difference (two-sided exact sign test, p = 1.0).